

- 2 The two-spot ladybird, *Adalia bipunctata*, is a species of small, flying beetle that is found in northern Europe and other parts of the world. The wings of these beetles are covered by two tough structures known as elytra, as shown in Fig. 2.1.

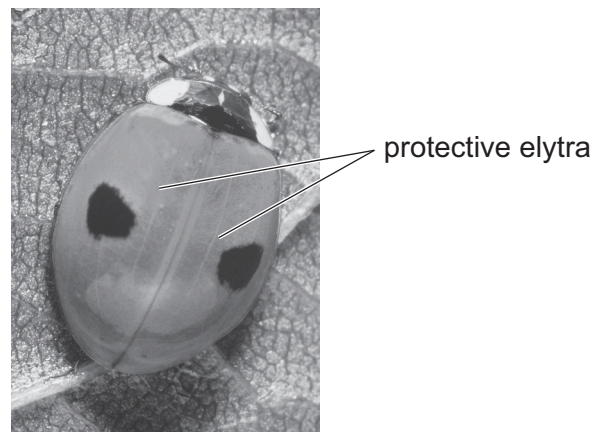


Fig. 2.1

A. bipunctata is an important predator of insect pests such as aphids. These insect pests feed on plants, including many crop species. Population numbers of *A. bipunctata* fluctuate in response to changes in the population numbers of their prey.

The elytra of *A. bipunctata* show phenotypic variation. The three most common morphs (forms) found in northern Europe are known as typica (**T**), quadrimaculata (**Q**) and sexpustulata (**S**).

- The typica morph, shown in Fig. 2.1, is mostly red with two black spots and is described as **non-melanic**.
- The other two morphs are described as **melanic** as they are mainly black with some red.
- The distribution of the colours in these three morphs is shown in Fig. 2.2.

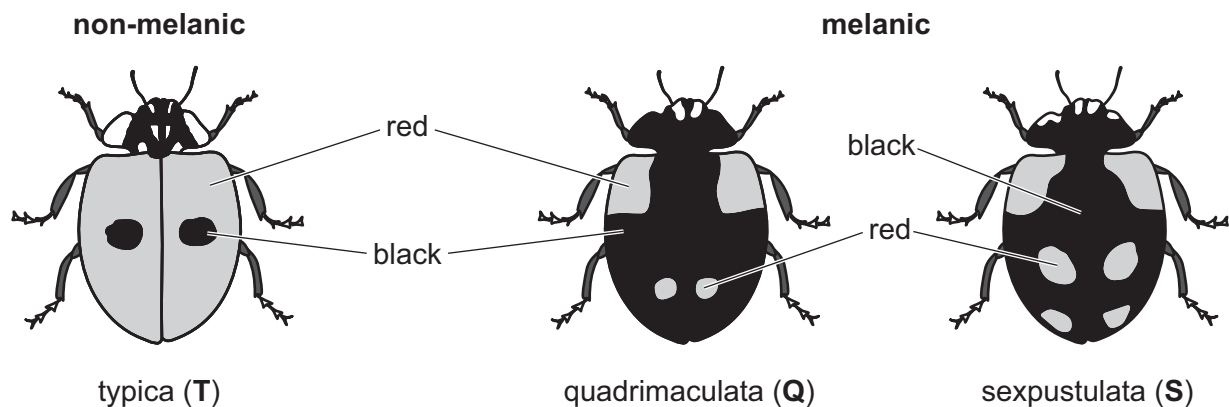


Fig. 2.2

In northern Europe, the populations of *A. bipunctata* hibernate (are inactive) during the winter. As the temperature increases in early spring, the populations become active. The populations remain active and produce three generations before the next winter begins.

- (a) It was observed that in some areas in populations of *A. bipunctata*, the proportion of non-melanic to melanic phenotypes in the early spring was different to the proportion later in the year in the autumn.

Researchers carried out a study in one area of Germany to compare the proportion of non-melanic to melanic phenotypes in early spring with the proportion in autumn after three generations had been produced. The data were collected over a period of 12 years.

The results are shown in Fig. 2.3.

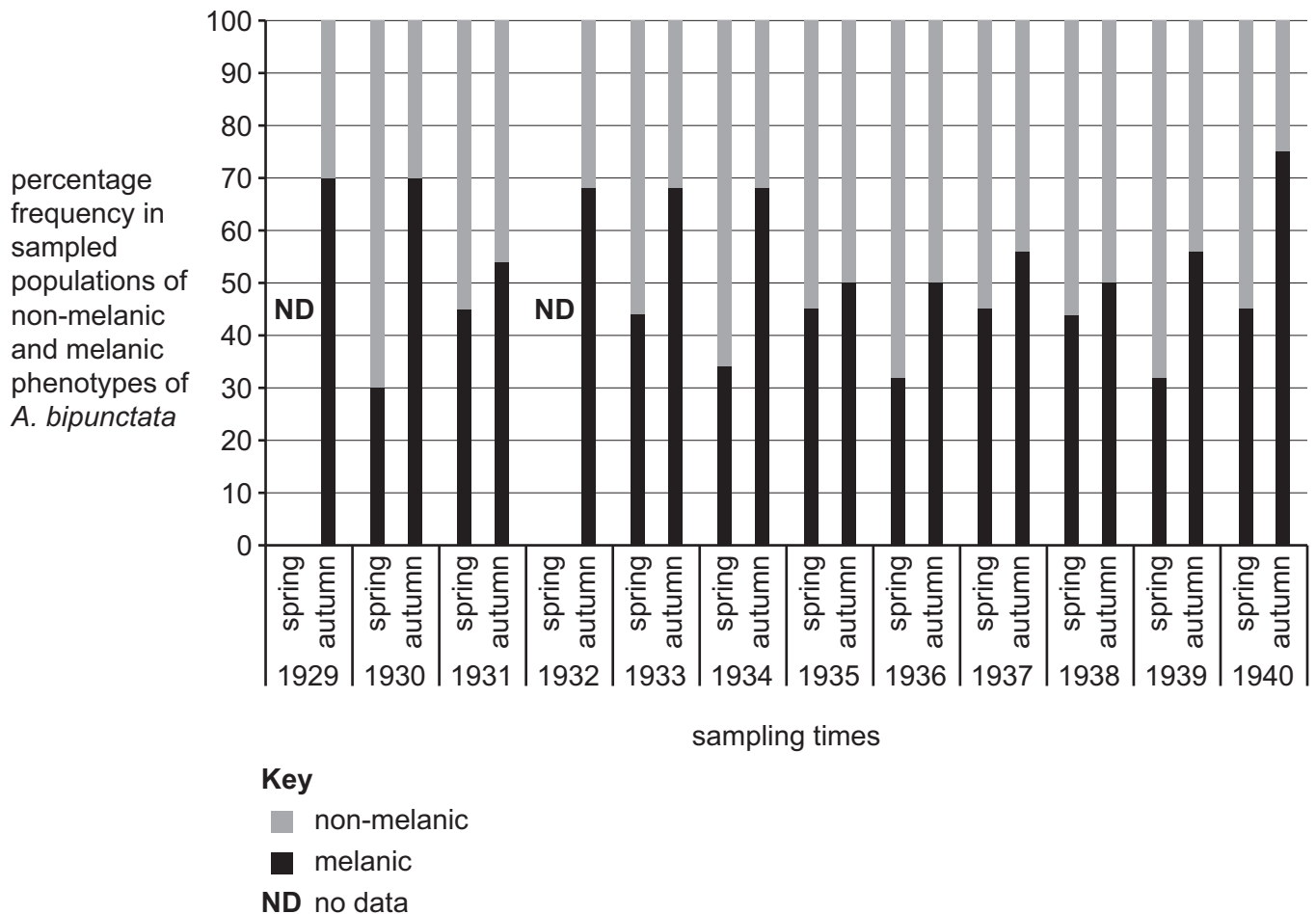


Fig. 2.3



- (i) Outline a method that could be followed to collect the data shown in Fig. 2.3.

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- (ii) The researchers concluded that there was a change in the proportion of the two phenotypes between spring and autumn, with the non-melanic phenotypes always showing a higher percentage frequency in the spring.

Suggest **two** pieces of extra information about the investigation that researchers should provide to improve confidence in their conclusion.

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- (b) Environmental temperature is one of many different selection pressures that could be acting on populations of *A. bipunctata*. The melanic phenotypes, morphs **Q** and **S**, are considered to be better adapted to the climate in northern Europe as they absorb heat energy more readily than the non-melanic phenotype, morph **T**.

Scientists investigated the effect of temperature on the body temperature of the two different phenotypes.

- The four individuals used in the investigation were collected while they were hibernating (inactive) and were kept outdoors for six weeks before the investigation.
- The individuals were weighed and arranged into pairs, each with approximately the same mass.
- A temperature sensor was fixed to the lower (ventral) surface of each individual.
- The temperature of each individual was recorded for 9 minutes. During that time, a heat-emitting lamp positioned directly above the individuals was switched on at 30 s and switched off at 5 minutes.

The results are shown in Fig. 2.4.

- (i) The researchers concluded that melanic morphs warm up more quickly than non-melanic morphs.

Explain how the information shown in Fig. 2.4 supports this conclusion.

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- (ii) State **two** further conclusions that can be made from the results shown in Fig. 2.4.

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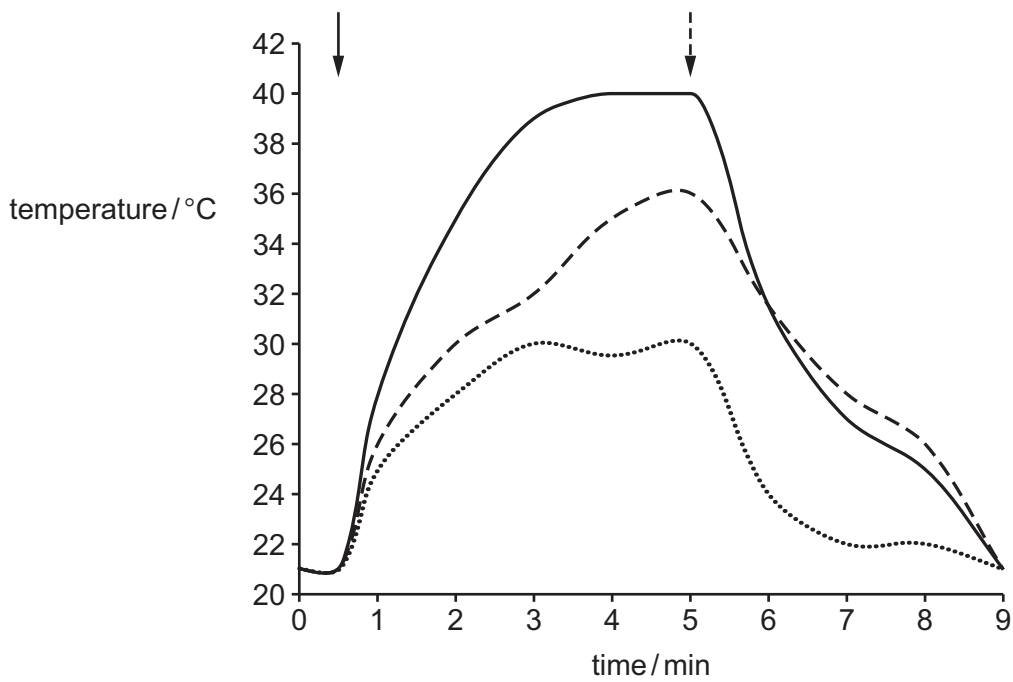
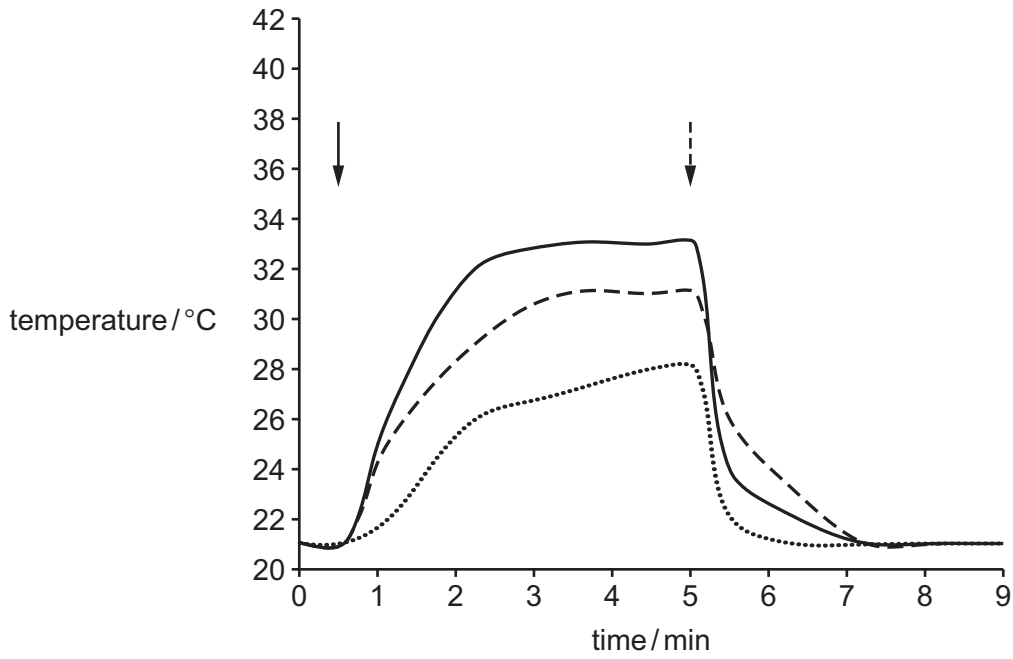
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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100



Key

- melanic phenotype
 - - - non-melanic phenotype
 - air temperature
- ↓ heat lamp on
- ↓ heat lamp off

Fig. 2.4

- (c) The scientists noticed that there appeared to be a relationship between body mass and the increase in body temperature. The scientists therefore continued to take results from another 17 individuals. They recorded the maximum difference between body temperature and the air temperature for each individual in their investigation.

All the results of the investigation are shown in Table 2.1 and Fig. 2.5.

Table 2.1

phenotype	morph	sex	mass/g	maximum temperature difference/°C
non-melanic	T	female	6.5	0.0
		male	8.4	1.1
		male	9.2	2.7
		male	10.0	2.9
		male	10.8	4.0
		female	11.5	5.4
		female	12.0	5.9
		female	13.1	5.3
		female	13.9	7.3
melanic	Q	male	8.4	3.5
		female	8.6	3.7
		male	10.1	5.3
		female	10.4	4.8
		female	11.0	8.5
		male	11.7	7.5
		female	13.1	7.3
		female	13.8	10.2
	S	male	8.6	4.1
		male	9.1	3.7
		male	10.8	5.9
		female	13.6	10.0

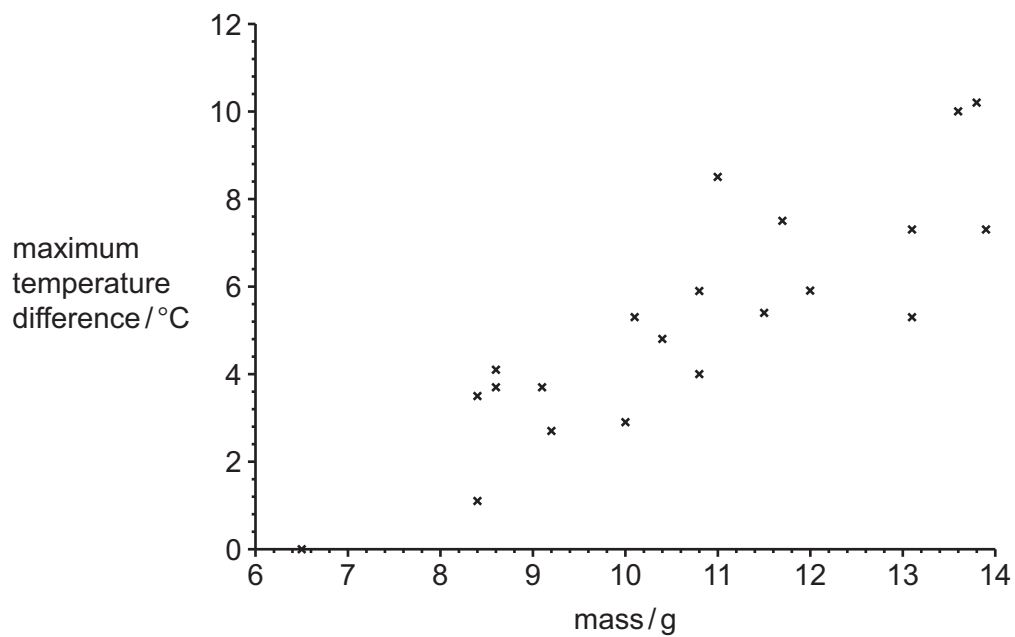


Fig. 2.5

State a statistical test that could be used to analyse the relationship shown in Table 2.1 and Fig. 2.5. Justify your choice.

statistical test

justification

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- (d) The researchers assumed that the appearance of the elytra was controlled by a single gene with two alleles. Breeding experiments showed that the melanic phenotype is dominant to the non-melanic phenotype.

Researchers made the hypothesis:

Some melanics are homozygous dominant, and some melanics are heterozygous.

Outline a breeding experiment that could be carried out to test this hypothesis, and state the results you would expect if the hypothesis is supported.

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[Total: 15]

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